MIL-B-50873A (MU)
28 July 1971
SUPERSEDES
MIL-B-50873 (MU)
19 February 1971

MILITARY SPECIFICATION

BAG, BARRIER, ELASTOMERIC

1. SCOPE

1.1 Scope.-This specification covers one type of elastomeric barrier bag intended for use with Cartridges, 152MM.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification and Marking of U.S. Military Property
MIL-STD-190	-	Identification and Marking of Rubber Products
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-417	-	Rubber Compositions, Vulcanized, General Purpose, Solid.
MIL-STD-1167	-	Ammunition Data Cards
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables For Inspection by Attributes
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of
MIL-A-2550	-	Ammunition and Special Weapons; General Specification For

FSC: 1320

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY

9260119 - Bag, Barrier, Elastomeric

PUBLICATIONS

ET-9260119 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extend specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 746	-	Brittleness Temperatures of Plastics and Elastomers by Impact
D 1149	-	Test for Accelerated Ozone Cracking of Vulcanized Rubber
D 518	-	Test for Resistance to Surface Cracking of Stretched Rubber Compounds (Method A)
D 685	-	Conditioning of Paper and Paper Products for Testing
E 96	-	Water Vapor Transmission of Materials in Sheet Form (Procedure E)

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

3. REQUIREMENTS

3.1 Materials.-The materials used in the manufacture of the barrier bag shall be an elastomeric compound which, when vulcanized, or cured, will comply with all the requirements specified in Table I and 3.3 through 3.14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I
Physical Properties

Property	Requirement	Test Paragraph
Tensile Strength	1900 psi Minimum (Min.)	4.3.3.1
Ultimate Elongation	700% Min.	4.3.3.2
Low Temperature Embrittlement	-50 Degrees F Maximum (Max.)	4.3.3.3
Tear Strength	160 pli Min.	4.3.3.4
Tension Set	26% Max.	4.3.3.5
Heat Aging		
Tensile Strength	1800 psi Min.	4.3.3.6
Ultimate Elongation	400% Min.	4.3.3.6
Oil Resistance		
Tensile Strength	480 psi Min.	4.3.3.9
Ultimate Elongation	180% Min.	4.3.3.9

3.2 Parts.-The barrier bag shall comply with all requirements contained in the applicable drawings and specifications.

3.3 Resistance to ignition.-The bag material shall not be ignited or melted through when tested in accordance with 4.3.3.7.

3.4 Ozone resistance.-The bag material shall not crack or check when tested in accordance with 4.3.3.8.

3.5 Compatibility.-The barrier bag material shall be compatible with the following materials for long term storage:

Steel (Types 52-100; ASTM-A-274, ASTM-A-322, and 43-40)

Aluminum (Type 7075-T-6)

Sintered Iron

Gilding Metal

Nonmetallic Case Material

Alloy of 90% Tin, 8% zinc and 2% Cadmium

Testing shall be conducted in accordance with 4.3.3.10 (see 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

3.6 Assembly and removal.-The barrier bag shall be of such a nature that it (a) can be assembled by hand (one man with no special tools) to a simulated 152MM cartridge with a minimum of difficulty; (b) will be formfitting to the cartridge case with no bulges or looseness; and (c) can be removed from a simulated cartridge with a pull of not less than fifteen (15) pounds and not more than fifty (50) pounds when tested in accordance with 4.3.3.11.1, and no less than 10 pounds or more than 115 pounds when tested in accordance with 4.3.3.11.2.

3.7 Blocking.-The barrier bag shall show no evidence of blocking when tested in accordance with 4.3.3.13.

3.8 Air pressure.-The barrier bag shall exhibit no air leakage when tested in accordance with 4.3.3.14.

3.9 Stability.-The stability of the nonmetallic cartridge case material shall not be decreased by the bag material when tested in accordance with 4.3.3.12 (see 6.5).

3.10 Workmanship.-The barrier bag shall be fabricated in a thorough, workmanlike manner in accordance with 4.3.2.1.

3.11 Moisture penetration.-Moisture penetration of the barrier bag shall be 2.5% maximum when tested in accordance with 4.3.3.15.

3.12 Stress relaxation.-The barrier bag material shall exhibit no stress relaxation, between -40°F and +40°F, when tested in accordance with 4.3.3.16.

3.13 Moisture vapor transmission rate (MVTR).-The MVTR of the barrier bag material shall be 0.35 gm/100 in^2/24 hrs. maximum when tested at 100 $\pm$ 2°F in accordance with 4.3.3.17.

3.14 First Article Inspection.-This specification makes provisions for first article inspections. Requirements for the submission of first article samples by the contractor shall be as specified in 4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to initiation of regular production, the contractor shall submit a first article sample to the Government facility designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of a minimum of twenty (20) barrier bags from each mold cavity intended for use in production or 150 barrier bags (see 6.5), whichever is greater, which have been produced by the contractor, or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment, which will be used in fulfilling the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

All parts and materials, including packaging and packing shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply so as to significantly affect product uniformity as determined by the Government).

4.2.2 Inspections to be performed.-The barrier bag will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, and any or all requirements of the applicable drawings. All tests shall be performed as specified.

4.2.3 Rejection.-If any part fails to comply with any of the applicable requirements for the inspections performed in accordance with 4.3.3, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part in the sample to comply with any or all of the stated requirements.

4.2.4 New equipment approval.-Prior to the use of a new or reworked mold cavity in regular production of barrier bags, the contractor shall submit to the Government a minimum of twenty (20) barrier bags produced from each new or reworked mold cavity for inspection. If any barrier bag fails to comply with the applicable drawings or specification requirements, when tested as specified in 4.2.2, the affected mold cavity shall not be approved for use in regular production. As a minimum, inspection shall include examination in accordance with 4.3.2.1 and drawing 9260119 and testing for moisture penetration in accordance with 4.3.3.15 (see 6.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the goups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4. Unless otherwise specified, general inspection Level II shall apply. Wall thickness shall be determined in accordance with FED-STD-601, Method 201.1 (see 6.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.2.1 Bag, Barrier, Elastomeric (see drawing 9260119) (see 6.7).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Thickness (see 6.8)	Gage	01001
102.	Gate(s) trimmed	Gage	01002
103.	Absence of talc	Visual	01008
104.	Creases, blisters, flow lines and voids on sealing rings (see 6.7)	Gage/ Visual	01009
105.	Blocking, holes and tears on any surface	Visual	01010
Minor:	AQL 1.50 percent		
201.	Blisters on base and sidewall, (see 6.7)	Gage	01011
202.	Flashing	Gage	01013
203.	Marking incorrect, illegible or missing	Visual	01015
204.	Oil, dirt or other foreign matter	Visual	01016

4.3.3 Testing

4.3.3.1 Tensile strength (see 3.1), Major defect, Code No. 02001.-One test specimen shall be prepared from each of 5 randomly selected barrier bags from each lot and tested in accordance with FED-STD-601, Method 4111 using Die III for specimen preparation. If the calculated median test value for the five specimens fails to comply with the applicable requirment, the lot shall be rejected (see 4.3.5).

4.3.3.2 Ultimate elongation (see 3.1), Major defect, Code No. 02002.-One test specimen shall be prepared from each of 5 randomly selected barrier bags per lot and tested in accordance with FED-STD-601, Method 4121 using Die III for specimen preparation. If the calculated median test value for the five specimens does not comply with the applicable requirement, the lot shall be rejected (see 4.3.5).

4.3.3.3 Low temperature embrittlement (see 3.1), Major defect, Code No. 03001.-Two specimens shall be prepared from

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

each of 5 barrier bags from the first article sample and tested for embrittlement in accordance with ASTM-D-746. The width of the specimen shall be .025 plus or minus .005 inches. Failure to comply with the applicable requirement shall be cause for rejection of the first article.

4.3.3.4 Tear strength (see 3.1), Major defect, Code No. 04001.-One test specimen from each of 7 randomly selected barrier bags per lot shall be prepared and tested in accordance with FED-STD-601, Method 4211 using Die C for specimen preparation. If the calculated median value for the seven specimens does not comply with the applicable requirement, the lot shall be rejected (see 4.3.5).

4.3.3.5 Tension set (see 3.1), Major defect, Code No. 05001.-One specimen from each of 5 randomly selected barrier bags per lot shall be prepared and tested in accordance with FED-STD-601, Method 4411. The elongation used for this test shall be 600 percent (7" from 1" length). Die III dumbbell specimens shall be used with a 1" bench mark. If the calculated median value for the 5 test specimens fails to comply with the applicable requirement, the lot shall be rejected. (see 4.3.5)

4.3.3.6 Heat aging (see 3.1), Major defect, Code No. 06001.-One test specimen from each of 5 barrier bags from the first article sample shall be prepared and tested in accordance with FED-STD-601, Method 7241. The specimens shall be heat aged at 212 plus or minus 2 degrees F for 70 plus or minus 1 hours. Failure of the specimens to comply with the applicable physical requirements after aging shall be cause for rejection of the first article sample.

4.3.3.7 Resistance to ignition (see 3.3), Major defect, Code No. 07001.-Three barrier bags from the first article sample shall be tested for resistance to ignition as specified in 4.4.6. Any evidence of burn-through of any bag specimen shall be cause for rejection of the first article sample.

4.3.3.8 Resistance to ozone (see 3.4), Major defect, Code No. 08001.-One specimen shall be prepared from each of five barrier bags from the first article sample and subjected to ozone resistance test in accordance with ASTM-D-1149, and ASTM-D-518, Method A. Each test specimen shall be elongated 12 1/2% for the test. Specimens shall be exposed to an ozone concentration of 50 parts per hundred million of air for 10 days at 100 ± 2 degrees F. If any sample cracks or checks in less than 10 days, the first article sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.9 Oil resistance (see 3.1), Major defect, Code No. 09001.-One specimen from each of 3 barrier bags from the first article sample shall be submerged in hydraulic fluid (MIL-H-6083) for 70 ± 1 hours at 125 ± 2°F. Samples shall then be prepared and tested in accordance with FED-STD-601, Method 6001 and 6111. Failure of the conditioned specimens to comply with the applicable requirements shall be cause for rejection of the first article sample.

4.3.3.10 Compatibility (see 3.5), Informational.-Two barrier bags shall be randomly selected from the first article sample and subjected to a compatibility test as specified in 4.4.7 for information.

4.3.3.11 Assembly and removal (see 3.6), Major defect, Ambient temperature Code No. 10001, Extreme temperature (125°F) Code No. 10002, Extreme temperature (-40°F) Code No. 10003.

4.3.3.11.1 Ambient temperature.-Fifty (50) barrier bags shall be randomly selected from each lot and subjected to an assembly-removal test in accordance with 4.4.1 and using test equipment as specified in 4.3.4. After 10 consecutive lots have been accepted, sampling shall be in accordance with Table II. If, during subsequent production, failure of a lot occurs, sampling shall revert to 50 barrier bags per lot until 10 consecutive lots have been accepted. If more than one barrier bag fails to comply with the requirements of 3.6a or 3.6b, the lot shall be rejected. If any barrier bag fails to comply with the requirement of 3.6c, the lot shall be rejected.

TABLE II

LOT SIZE	NO. OF SAMPLES
up to 1599	10
1600 to 1999	20
2000 to 2399	30
2400 to 2799	40
2800 and over	50

4.3.3.11.2 Extreme temperatures.-From the first article sample, ten barrier bags shall be selected and subjected to the assembly-removal test in accordance with 4.4.1 after conditioning the test assembly with barrier bag for a minimum of 4 hours, 5 samples at -40 degrees F, and 5 samples at 125 degrees F. The first article sample shall be rejected if any sample fails to comply with the applicable requirement of 3.6c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.12 Stability (see 3.9), Major defect, Heat test, Code No. 11001, Vacuum test Code No. 11002.

4.3.3.12.1 Heat test (134.5 degrees C).-A specimen weighing approximately 1.25 grams shall be prepared from a first article sample barrier bag and together with 1.25 grams of nonmetallic cartridge case material, shall be tested in accordance with MIL-STD-286A, Method 404.1.2. The time required to change the methyl violet paper to salmon pink shall be noted and observation for explosion prior to 300 minutes of testing shall be made (see 6.5). Change to a salmon pink within 35 minutes or explosion within 300 minutes shall be cause for rejection of the first article sample.

4.3.3.12.2 Vacuum test.-Two 2.5 gram specimens shall be prepared from a first article sample barrier bag and tested in accordance with MIL-STD-286A. Method 403.1.2, with the following modification. The test specimen shall consist of 2.5g of the barrier bag material and 2.5g of nonmetallic cartridge case material. A control sample of 2.5g of case material and a control sample of 2.5g of bag material shall also be tested. The gas evolution of the two controls are added and the sum is subtracted from the gas volume obtained from the test mixture. If the calculated difference is greater than 5.0 ml, the first article sample shall be rejected.

4.3.3.13 Blocking (see 3.7), Considerable blocking, Major defect, Code No. 12001, Slight blocking, Minor defect, Code No. 12002.-Six barrier bags shall be selected at random from each lot, except that sample size shall be three for each lot of 1600 units or less, and subjected to a blocking test in accordance with 4.4.2, and using test equipment specified in 4.3.4. If one or more samples show signs of considerable blocking or two or more samples show signs of slight blocking, the lot shall be rejected (see 6.6).

4.3.3.14 Air pressure test (see 3.8), Major defect, Code No. 13001.-Fifty barrier bags shall be selected at random from each lot, and subjected to an air pressure test in accordance with 4.4.3, using test equipment as specified in 4.3.4. After 10 consecutive lots have been accepted, sampling shall be in accordance with Table II (see 4.3.3.11.1). If a lot is rejected during subsequent production, sampling shall revert to 50 barrier bags per lot until 10 consecutive lots have been accepted. If any sample shows evidence of air leakage through the base, sidewall, or sealing rings, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.15 Moisture penetration (see 3.11), Major defect, Code No. 14001.-Ten (10 barrier bags from each mold cavity shall be selected from the frist article sample and subjected to a moisture penetration test as specified in 4.4.4, using test equipment specified in 4.3.4. If any silica gel bag increases in weight in excess of 2.5 percent in 24 hours the first article shall be rejected. All samples shall be conditioned at 23 ± 2°C and 50 ± 2% R.H. for 24 hours minimum prior to testing.

4.3.3.16 Stress relaxation (see 3.12), Major defect, Code No. 15001.-One specimen from each of two barrier bags in the first article sample shall be tested for stress relaxation as specified in 4.4.5. If any specimen exhibits relaxation in the specified temperature range, the first article sample shall be rejected.

4.3.3.17 Moisture vapor transmission rate (MVTR) (See 3.13), Major defect, Code No. 16001.-Duplicate specimens from each of two barrier bags in the first article sample shall be tested for MVTR in accordance with ASTM-E96 procedure E or using a Honeywell-St. Regis MVTR apparatus or equivalent. Results shall be reported in gm/24 hrs/100 in^2. If any specimen fails to meet the applicable requirement, the first article sample shall be rejected.

4.3.4 Inspection equipment.-Index of Inspection Equipment List Number ET-9260119 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification or stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipmeht listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.5 Small lot sampling.-For production lots of 1600 units or less, sampling for tensile strength, ultimate elongation, tear strength and tension set may be reduced to one specimen from each of 3 barrier bags for each test. If one or more specimens fails the applicable requirement, testing shall revert to that specified in the applicable paragraphs above until three consecutive lots have been accepted.

4.4 Test methods

4.4.1 Assembly removal.-The barrier bag shall be assembled to the test fixture by hand (one man, no special tools). Any difficulty in assembling the bag shall be recorded. The bag shall be examined for compliance with 3.6b. The bag shall be removed from the test fixture in accordance with the following procedure or a Government approved equivalent procedure.

a. The assembly shall be held firmly in place by clamping the forward end of the assembly in a suitable holding fixture.

b. A suitable means of applying a pull force, such as a snap clamp, shall be affixed to one bag flap.

c. The clamp shall be connected to a pressure-actuated piston by straps or wires.

d. The pressure-actuated piston shall be positioned directly to the rear of the assembly and in such a manner that the piston movement will be along the longitudinal axis of the assembly.

e. A rearward force shall be imparted to the piston at a rate of 20 inches per minute until the bag is completely removed from the assembly. The maximum force required to remove each bag shall be recorded. Samples that have been conditioned at extreme temperatures shall be tested as above while contained in the conditioning chamber.

4.4.2 Blocking.-From each barrier bag, two 4" x 4" test specimens shall be cut from the smooth area of the side wall. (Assure that no dusting powder exists on any surface of the test specimen.) The two specimens, outer face to outer face, shall be placed between the glass plates and subjected to an even force of 20 $\pm$ 1 pound for 24 $\pm$ 1 hours at 95 $\pm$ 5 degrees F and 95 $\pm$ 5% R.H. The specimens, glass plates and weight shall be removed from the conditioning chamber and allowed to cool for 5 $\pm$ 2 minutes at room temperature. The weight and glass plates shall be removed and the mated surfaces of the two test specimens shall be examined for signs of blocking. First article samples shall be tested as above, except that conditioning time shall be for 120 $\pm$ 1 hours. The test conditions (temperature and relative humidity) and degree of blocking (see 6.6) shall be recorded for each sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.3 Air pressure.-The barrier bag shall be clamped to the test fixture such that no more than the first two sealing rings at the open end of the bag are covered by the clamp. The clamp shall be attached securely to prevent the escape of air through the clamped sealing rings. Air shall be introduced at 0.75 ± 0.25 psi. While maintaining the specified air pressure, the pressurized barrier bag shall be immersed in water, base end down, until the water level reaches the edge of the clamp, for approximately 30 seconds. Observation shall be made for evidence of air escaping (bubbles) through the base end, sidewall and the sealing rings of the barrier bag.

4.4.4 Moisture penetration.-Bagged silica gel (28 ± 2 grams silica gel) shall be conditioned for a minimum of 2 hours at 105 ± 2 degrees C. The dry silica gel shall be weighed, the weight recorded, and then placed inside the perforated metal case. The case shall then be assembled to the projectile. The barrier bag shall be assembled to the projectile such that the bag extends over the rotating band of the projectile and the flaps of the bag are not folded. The assembly shall then be placed in a temperature-humidity chamber in a horizontal position and subjected to the following conditioning cycle at constant 100% relative humidity for 24 hours:

a. 4 ± 1/4 hours at 70 ± 5 degrees F
b. 2 ± 1/4 hours transition stage
c. 16 ± 1/4 hours at 105 ± 5 degrees F
d. 2 ± 1/4 hours transition to 70 ± 5 degrees F

The assembly shall be removed from the chamber and the barrier bag and case shall be disassembled. Within 5 minutes of removal from the chamber, the bagged silica gel shall be weighed. The percent moisture pickup shall be calculated by the following equation:

$$\text{Percent moisture pickup} = [(W_F - W_I) / W_I] \times 100$$

Where:

W_F = Final silica gel weight

W_I = Initial silica gel weight

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.5 Stress relaxation.-Place a 1 inch by 5 inch barrier bag specimen in a tensile tester with the jaws of the grips 1 inch apart. Impart an elongation of 200% to the specimen. Maintain the elongation constant throughout the test. Reduce the specimen temperature in the grips to the test temperature. Test temperatures shall be between -40°F and +40°F in increments of not less than 10°F and not more than 20°F. Any given test temperature shall be maintained at ± 2°F for a minimum of 40 hours. When at equilibrium temperature, record the stress level. Every 2 hours for the first 10 hours and every 5 hours thereafter, record the stress level. Prepare a graph of stress level (either psi or total load) on the y-axis and logarithm of time on the x-axis. If a definite downturn in the curve is noted, the specimen has relaxed. No relaxation is indicated by a curve of constant zero slope. Repeat at other selected temperatures.

4.4.6 Resistance to ignition.-Cut out the cylindrical portion of the sidewall of each barrier bag (see dwg. 9260119).- Prepare a sandwich consisting of a bag sample and a 2" x 2" piece of cartridge case material on each side of the bag sample. The pieces of case material are to be opposite each other on the two sides of the bag sample. Do not use adhesive or tape to secure the case material to the bag. Assemble each of the three test samples on a clean, dry, non-metallic, non-combustible surface. Ignite the upper piece of cartridge case material with a match and then immediately remove the match. Allow sufficient time for both pieces of cartridge case material to burn out. Examine each bag specimen for burn-through.

4.4.7 Compatibility.-Assemble a 152MM XM657 projectile to a nonmetallic cartridge case using a locking ring only. Place the outer cup of an ignition element, electrical (dwg. 9249028) into the initiator boss. Assemble one barrier bag over the cartridge case with the bag flaps extending over the rotating band of the projectile. Repeat the above assembly for the other sample barrier bag using a 152MM XM625 canister. Condition the assemblies at 160 ± 2°F and 95 + 5% R.H. for 166 + 1 hours. Allow the assemblies to return to ambient conditions. Disassemble and inspect the projectile cartridge case, ignition element cup and barrier bag for deterioration; in general, evidence of staining and corrosion of the metal parts and softening of the cartridge case. One copy of the test results shall be forwarded to Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-ND2 (see 6.5).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.-The material shall be prepared for delivery in accordance with the best commercial practice in such a manner as to insure acceptance at the first delivery station.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

5.2 Marking.-Shipments shall be marked in accordance with Standard MIL-STD-129. Components shall be marked in accordance with the applicable drawing, MIL-STD-190 and MIL-STD-130.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU) and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate, and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Provisions for submission of first article sample.- Instructions as to the location for the evaluation of first article sample shall be obtained from the contracting officer. Upon receipt of such requests, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries shall be forwarded to:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NK1
Dover, New Jersey 07801

6.5 Testing responsibility.-The contractor is required to perform all testing and examinations contained in the specification with the exception that the resistance to ignition, compatibility, moisture penetration, and stability tests specified in 4.3.3.7, 4.3.3.10, 4.3.3.15 and 4.3.3.12) shall be performed by the Government at Government expense.

6.6 Degree of blocking.-The degree of blocking shall be defined as follows:

a. No blocking.-No adhesion or cohesion between adjacent surfaces which slide freely upon one another. Surfaces of specimens are not marred when sheets are separated.

b. Slight blocking.-Slight adhesion; adjacent surfaces do not slide freely, and upon separation of the sheets, the surfaces show evidence of slight marring.

c. Considerable blocking.-Adhesion or cohesion of adjacent surfaces will be definitely marred or partially destroyed upon separation.

6.7 Visual defects

a. Creases inside the sealing area are acceptable, if by stretching or rubbing, the crease tends to disappear. Creases outside the sealing area are acceptable.

b. Blisters inside the sealing area not acceptable. Blisters not farther than three inches from the base of the bag are acceptable if no larger than 3/32 inch, no more than 5 per bag, and no more than 3 per square inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

6.8 Wall thickness.-Wall thickness shall be reported on four readings 90° apart midway between the sealing area and the base end of the bag, on one reading from each of the bag flaps, and on two readings on the base end of the bag taken approximately one inch from the edge of the base end 180° apart. Wall thickness shall be considered acceptable providing each of the eight readings meet the requirements of the applicable drawing.

6.9 Dimensional examination.-A minimum of 10 barrier bags from each mold cavity in the first article sample shall be examined for all drawing dimensions. Failure of any sample to meet all dimensional requirements will constitute disapproval of the affected mold cavity. These requirements must also be met for any new or reworked mold cavity intended for use in production following acceptance of the first article sample.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1320

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.